3. The equilibrium attained during the melting of ice is known as _____.
4. The pH of a fruit juice is 5.6. If you add slaked lime to this juice, its pH _____ (increse/decrese)
5. The value of ionic product of water at $25^0$ C is _____.
6. The normal pH of human blood is _____
7. Electrolysis is type of _____ reaction
8. The number of products formed in a synthesis reaction is _____
9. Chemical volcano is an example for _____ type of reaction
10. The ion formed by dissolution of $H^+$ in water is called _____

## III. Match the following

**1.** Identify the types of reaction

| REACTION | TYPE |
|---|---|
| $NH_4OH_{(aq)} + CH_3COOH_{(aq)} \rightarrow CH_3COONH_{4(aq)} + H_2O_{(l)}$ | Single Displacement |
| $Zn_{(s)} + CuSO_{4(aq)} \rightarrow ZnSO_{4(aq)} + Cu_{(s)}$ | Combustion |
| $ZnCO_{3(s)} \xrightarrow{Heat} ZnO_{(s)} + CO_{2(g)}$ | Neutralisation |
| $C_2H_{4(g)} + 4O_{2(g)} \rightarrow 2CO_{2(g)} + 2H_2O_{(g)} + Heat$ | Thermal decomposition |

## IV. True or False: (If false give the correct statement)

1. Silver metal can displace hydrogen gas from nitric acid.
2. The pH of rain water containing dissolved gases like $SO_3$, $CO_2$, $NO_2$ will be less than 7.
3. At the equilibrium of a reversible reaction, the concentration of the reactants and the products will be equal.
4. Periodical removal of one of the products of a reversible reaction increases the yield.
5. On dipping a pH paper in a solution, it turns into yellow. Then the solution is basic.

## V. Short answer questions:

1. When an aqueous solution of potassium chloride is added to an aqueous solution of silver nitrate, a white precipitate is formed. Give the chemical equation of this reaction.
2. Why does the reaction rate of a reaction increase on raising the temperature?
3. Define combination reaction. Give one example for an exothermic combination reaction.
4. Differentiate reversible and irreversible reactions

## VI. Answer in detail

1. What are called thermolysis reactions?
2. Explain the types of double displacement reactions with examples.
3. Explain the factors influencing the rate of a reaction
4. How does pH play an important role in everyday life?
5. What is a chemical equilibrium? What are its characteristics?

## VII. HOT questions

1. A solid compound 'A' decomposes on heating into 'B' and a gas 'C'. On passing the gas 'C' through water, it becomes acidic. Identify A, B and C.



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